is stationary at $\phi = \phi_0$:

$$0 = \left.\frac{\partial V}{\partial \phi_n}\right|_{\phi=\phi_0} = -\sum_m \mathscr{M}_{nm}\mathscr{F}_{m0} + \sum_A D_{A0}(\phi_0^\dagger t_A)_n \,. \tag{27.5.13}$$

We also need the gauge invariance condition (27.4.12), which at $\phi = \phi_0$ reads

$$\sum_n \mathscr{F}_{n0}\,(t_A\phi_0)_n = 0\,. \tag{27.5.14}$$

Combining Eqs. (27.5.13) and (27.5.14) with Eqs. (27.5.5) and (27.5.6) then yields

$$i\sqrt{2}\sum_m (M^\dagger M)_{nm}\mathscr{F}_{m0} = i\sqrt{2}\sum_A D_A(t_A\mathscr{F}_0^*)_n = -\sum_A (M^\dagger M)_{nA}D_{A0} \tag{27.5.15}$$

and

$$i\sqrt{2}\sum_m (M^\dagger M)_{Am}\mathscr{F}_{m0} = -2\sum_{nm}\mathscr{F}_{n0}\,(t_A)_{nm}\,\mathscr{F}_{m0} = -\sum_B (M^\dagger M)_{AB}\,D_{B0}\,. \tag{27.5.16}$$

That is,

$$M^\dagger M \begin{pmatrix} i\sqrt{2}\mathscr{F}_0 \\ D_0 \end{pmatrix} = 0\,, \tag{27.5.17}$$

as was to be shown.

27.6 Perturbative Non-Renormalization Theorems

From the beginning, several of the ultraviolet divergences in ordinary renormalizable quantum field theories were found to be absent in the supersymmetric versions of these theories. With the development in 1975 of supergraph techniques, in which all particles in each supermultiplet are considered together, it became possible to show that some radiative corrections are not only finite, but are absent altogether in perturbation theory.[5] Supergraphs will be described in detail in Chapter 30, but as it happens they are not needed to prove the most important non-renormalization theorems. This section will give a version of a method developed by Seiberg[6] in 1993, which showed how the non-renormalization theorems may be easily obtained from simple considerations of symmetry and analyticity.

Consider a general renormalizable supersymmetric gauge theory with a number of left-chiral superfields Φ_n and/or gauge superfields V_A. As mentioned in Section 27.3, if we remove the factor g in the t_A and C_{ABC} and include it instead in the gauge superfields, then the Lagrangian density